EXPERIMENTS AND THEIR INFERENCE

EXP 1: BASELINE

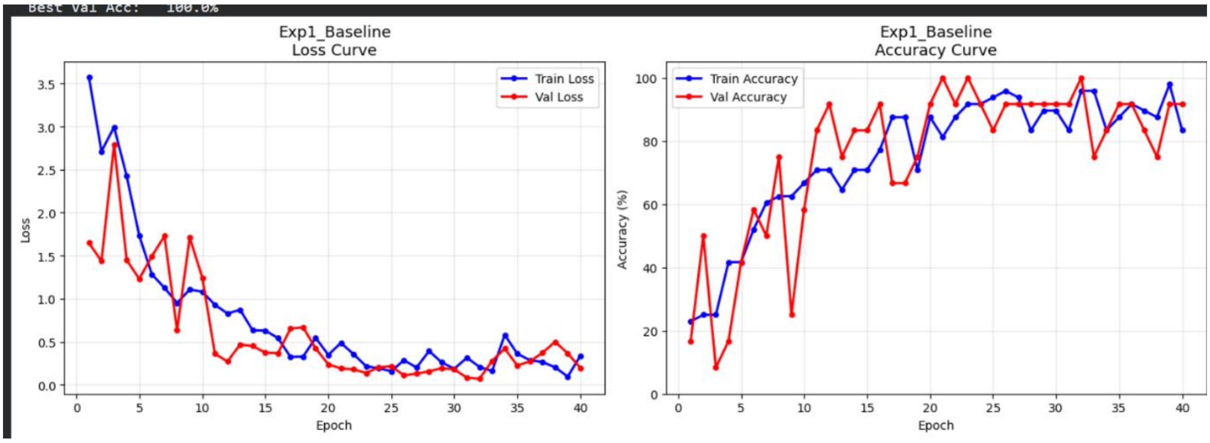
All the parameters are in their default values.

Adam optimizer automatically adjusts the learning rate.(LR=.001 usually used for adam)

Dropout is 0.5 which means that 50 percent of neurons will be turned off during training.

Batch size is 8 as we have 48 images in the data set.

The loss curve for this case is:

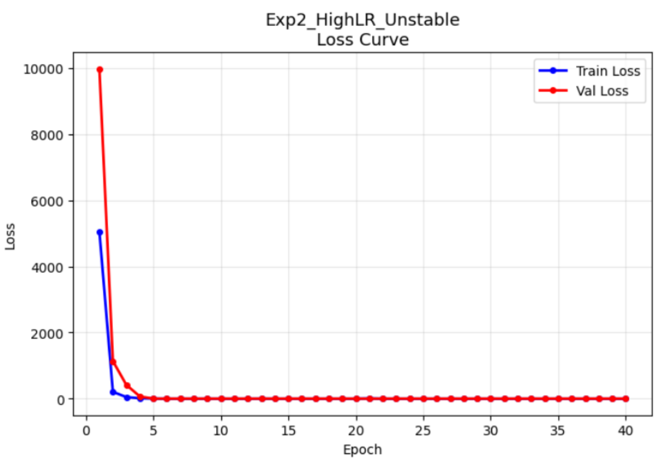


OBSERVATION:
We observe that the loss decreases as the epoch increases. Also the gap between the training and validation loss is less.

EXP 2: HIGH LEARNING RATE

Here I tried altering the learning rate to check and compare it with the baseline case . The LR is increased by setting it to 0.1 from 0.001.

Loss curve for this case:



OBSERVATION:

We see that the graph suddenly descents till epoch 5 and then flattens out. We also see that the loss suddenly spiked up to 1000 in this case from 3.5 in the last case.

INFERENCE:

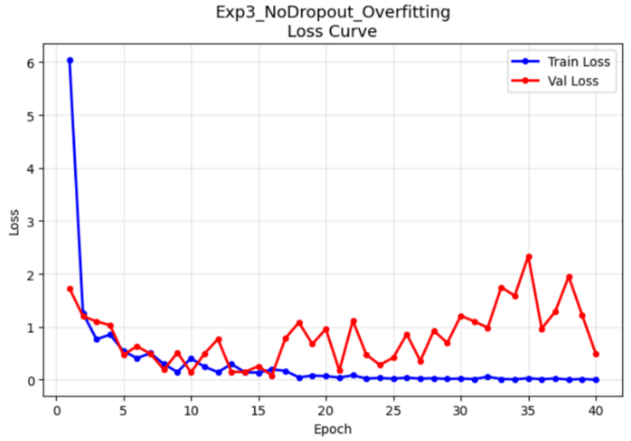
This graph shows different results on increasing the LR. Usually when LR is increased, the loss curve is bouncing (up and down) due to the large step size .

However, here we notice that the spiked loss descends and then flattens out. The loss spiked due to huge updates and changes in weights and gradients. This is called Exploding gradients. Then the slope flattens out due to the effect of Adam optimizer that keeps a track of the previous gradients. This also causes overfitting in the model.

EXP 3: NO DROPOUT

Here, the dropout is set to 0, which means all the neurons are active while training.

The loss curve for this case is:



OBSERVATION:

The loss curve for training loss is continuously decreasing, however, the loss curve for validation is bouncing after 15. The gap between training and validation loss is greater than the baseline case.

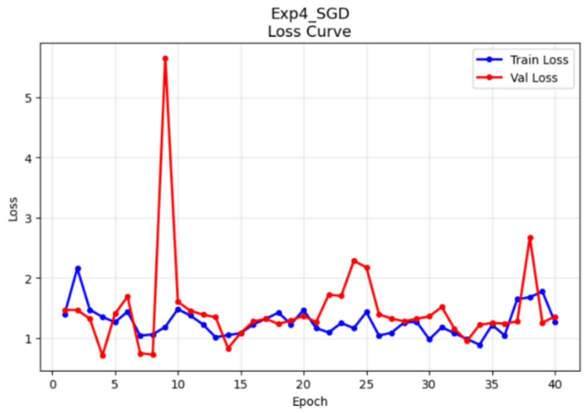
INFERENCE:

due to 0 dropout, all the neurons are active during training, this means that all the neurons have seen all the training images. This causes Overfitting, meaning that the model starts to memorize the images and instead to learning how to generalize. Thus, it is recommended that the dropout should not be 0 .

EXP 4: SGD OPTIMIZER

The Adam optimizer is changed to SGD optimizer. The SGD moves in the direction of decreasing descent by a fixed step.

The loss curve here is:



OBSERVATION:

Random spikes in val loss. The training loss first decreases and then increase again towards the end.

INFERENCE:

From the graph we infer that the model does not really learn the dataset completely, there are bouncing curves, this mainly happens because the SGD does not have have the memory of the past gradients.

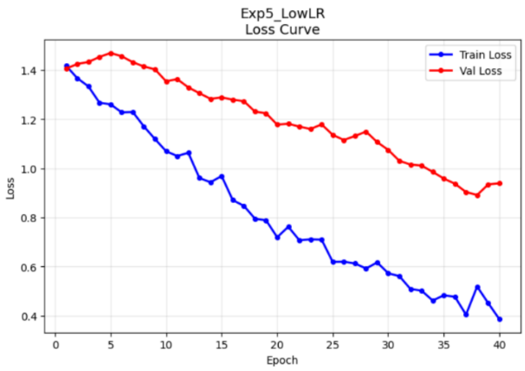
However, we see that the training and validation loss have close by values, this means that the model is not overfitting but its not leaning anything either.

Thus, adam works better when compared to SGD. the curves in the first baseline case where Adam is used are smooth as compared to slopes of SGD.

EXP 5: LOW LEARING RATE

Setting the learning rate to 0.0001 lowering it by 100 times.

The loss curve for this case is:



OBSERVATION:

Both the losses decrease. they have a huge gap as the epochs progress.

INFERENCE:

We see that the slopes are continuously decreasing, this means that the model is still training and has not finished training. In this case, underfitting occurs meaning the model has not learnt enough to predict the output. Due to reducing the LR , the step size gets reduced , thus the model will require a few more epochs to completely train. When the model trains, completely the slope flattens which shows has model has completed its training. Thus, reducing the LR will slow down the process.

COMPLETE SUMMARY OF THE EXPERIMENTS:

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                       EXPERIMENT SUMMARY
=================================================================
Experiment            Best Train Acc  Best Val Acc  Final Val Loss
Baseline                       97.9%        100.0%          0.1983
High LR                        45.8%         33.3%          1.5230
No Dropout                    100.0%        100.0%          0.4997
SGD                            60.4%         83.3%          1.3576
Low LR                         93.8%         50.0%          0.9390
=================================================================
```